Dynamic Approval Matrix Management

Last updated by | m.calderon.santiago@accenture.com | 21 abr 2026 at 11:30 CEST

Contents

- Dynamic Approval Matrix Management
  - Approval Matrix
    - Record Types: Opportunity, Order Request.
  - Automation Architecture & Orchestration
    - The Assignment Layer (Record-Triggered Flows)
    - The Execution Layer (Standard Approval Process)
  - Scalability & Group Strategy
  - Architectural Benefits summary
  - Approval Matrix Validation & Exception Handling
    - Zero-Entry Exception Logic
    - Error Message Specification
    - Architectural Benefits

Dynamic Approval Matrix Management

This solution shifts the complexity from rigid Standard Approval Processes to a data-driven configuration model. By utilizing a "Configuration Object" (`Approval Matrix`), business administrators can manage thresholds and approvers dynamically without requiring system administrator intervention or deployment of new metadata.

Approval Matrix

This object acts as the central "Brain" for routing logic.

Field Label	Data Type	Logic / Functional Behavior
Target Object	Picklist	Defines the application scope (e.g., *Opportunity, Order Request*).
Sales Organization	Global Picklist	The specific geographic or structural branch the rule applies to.
Business Unit	Global Picklist	The specific business line the rule applies to.
Threshold L2	Currency (16,2)	Minimum transaction amount to trigger the 2nd level of approval.
Threshold L3	Currency (16,2)	Minimum transaction amount to trigger the 3rd level of approval.
Approver L1	Lookup (Group)	Base level approver (Queue).
Approver L2	Lookup (Group)	Escalation approver for Level 2.
Approver L3	Lookup (Group)	Final escalation approver for Level 3.

Record Types: Opportunity, Order Request.

We will have one Record Type per sObject (Opportunity and Order Request) to make it easier to assign the sObject to the default value based on the recordType.

Automation Architecture & Orchestration

The routing mechanism relies on a "Stamp and Evaluate" pattern, split into two layers:

The Assignment Layer (Record-Triggered Flows)

For the initial rollout, two independent **Before-Save Record-Triggered Flows** will be deployed (one for `Opportunity`, one for `Order Request`).

- **Trigger:** Record creation or specific updates (e.g., changes to Amount or Status).
- **Query Logic:** The Flow queries the `Approval Matrix` where `Target Object`, `Sales Organization`, and `Business Unit` match the target record.
- **Stamping Action:** The Flow extracts the Matrix values and copies them into **hidden technical fields** on the target record (e.g., `Approver L1`, `Threshold L2`, etc.). We will also assign the Approval Matrix record to a lookup to have it.

The Execution Layer (Standard Approval Process)

A single, streamlined Standard Approval Process is built per object, utilizing the stamped fields for its entry criteria and routing:

- **Step 1:** Always executes. Assigned to the stamped `Approver L1`.
- **Step 2:** Conditional Step. Executes ONLY IF `Record.Amount > Record.Threshold L2`. Assigned to `Approver L2`.
- **Step 3:** Conditional Step. Executes ONLY IF `Record.Amount > Record.Threshold L3`. Assigned to `Approver L3`.

Scalability & Group Strategy

To ensure long-term maintainability and avoid bottlenecks (e.g., an approver going on vacation), the assignment architecture relies heavily on **Queues**:

- **Group Approvals:** For regions/departments with multiple valid approvers, a Queue is created and referenced in the Matrix. Any member of the Queue can claim and approve the record.
- **Single Approvers:** Even for regions with a single designated approver, it is highly recommended to assign a "Single-Member Queue" rather than the User directly. This allows admins to swap the user in the Queue during absences without modifying the Matrix data.

Architectural Benefits summary

- **Delegated Maintenance:** Business Managers can update routing and financial thresholds as easily as editing an Approval Matrix record.
- **Scalability:** Adding a new Business Unit or Sales Org simply requires creating one new Matrix record, rather than modifying Flow logic.
- **Auditability:** Standard Field History Tracking on the `Approval Matrix` object provides a clear compliance trail of *who* changed approval thresholds and *when*.

Approval Matrix Validation & Exception Handling

To guarantee that no transaction bypasses the necessary oversight, the system implements a **Strict Routing Policy**. If the automated assignment logic fails to find a valid rule in the `Approval Matrix`, the system will block the record's progress and alert the user.

Zero-Entry Exception Logic

When a record (Opportunity, Order Request) is created or its commercial criteria (Amount, Business Unit, Sales Org) are updated, the **Assignment Flow** performs a lookup. If this lookup returns no results (Null), the following "Guardrail" is triggered:

- **Mechanism:** Flow **Custom Error** Element.
- **Trigger Condition:** `Get_Approval_Matrix_Record` IS NULL.
- **System Action:** Immediate rollback of the DML operation (the record is not saved or updated).

Error Message Specification

The user will see a standardized error message on their screen to guide them toward a resolution:

> **Error Code: [AP-001] Missing Approval Mapping** *"No valid Approval Matrix was found for the current combination of Sales Organization and Business Unit. Please contact your System Administrator to define the required approvers before proceeding."*

Architectural Benefits

- **Audit Integrity:** Ensures 100% of the transactions are audited by the correct authority.
- **Data Quality:** Forces users and administrators to keep the organizational structure (Sales Orgs and BUs) synchronized between Salesforce and the business reality.
- **Risk Mitigation:** Prevents "ghost approvals" where a record might be approved by a default user who doesn't have the appropriate local context or financial authority.